Algebra 1 Honors Unit 13 Assessment Guide

General Information	Unit 13 was written so that students apply knowledge from prior benchmarks. Throughout the course, students have explored linear, quadratic, and exponential functions using both mathematical and real-world contexts. The benchmarks allow the teacher to cover content that will help students review previously learned material. This will help the teacher determine any weaknesses students may have and to help support student learning as students prepare for the end-of-course exam. Students were introduced to parent functions in unit 1. Students bring that introduction full circle by looking at parent functions through transformations. For many of these functions, students have seen transformations but have not named them as such. The formal understanding of transformations is further secured using multiple representations. Assessing both x- and y-intercepts in question 1 was purposefully done so that teachers can determine if students distinguish between the two and can determine each from multiple representations. The assessment aims to balance student understanding of all of the function types students have encountered throughout Algebra 1. Teachers may wish to determine if their students are struggling with a certain function type. There are two honors benchmarks in this unit. Teachers may wish to determine if students should be given formulas from the reference sheet. They were not provided because students are not writing functions. Question 5 does provide the formulas for simple and compound interest within the options.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1	Benchmark(s)	MA.912.F.1.6

	x –intercept at $(-3,0)$	x –intercept at $(1,0)$	y –intercept at $(0,3)$	y –intercept at $(0,4)$
Function B	□	□	□	■
Function F	■	□	□	■
Function H	□	□	□	■
Function M	■	■	■	□
Function R	□	□	□	■
Function W	□	■	■	□

Recommended Scoring (30 Total Points)	Consider giving students 3 points for each correct answer. Consider a deduction of 2 or 3 points for each incorrect selection. Students who select all of the options should not receive any points.

Question	2a	Benchmark(s)	MA.912.AR.1.1

a. Complete the statements.

The function $T(x) = 0.29(1.14)^x$ represents

- ○ linear decay.
- ○ linear growth.
- ○ exponential decay.
- ○ exponential growth.

The value 0.29 represents the

- ○ decay rate.
- ○ growth rate.
- ○ the largest size of the tumor, in cm^3.
- ○ the original size of the tumor, in cm^3.

Recommended Scoring (6 Total Points)	Consider 3 points for each statement. Partial credit is not recommended.

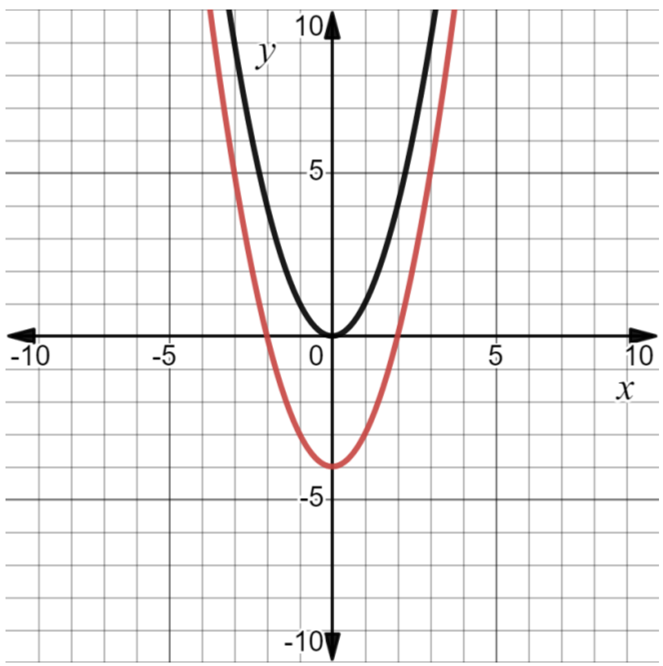
Question	2b	Benchmark(s)	MA.912.F.1.3
	$0.062\ \frac{cm^3}{day}$	**Recommended Scoring** **(4 Total Points)**	Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider a minor deduction of points.

Question	3a	Benchmark(s)	MA.912.F.1.1
	○ $y = x$ ○ $y = x^2$ ○ $y = x^3$ ○ $y = \sqrt{x}$ ○ $y = \lvert x \rvert$ ○ $y = 2^x$ ○ $y = \frac{1}{2}^x$	**Recommended Scoring** **(3 Total Points)**	Partial credit is not recommended.

Question	3b	Benchmark(s)	MA.912.F.1.1
	○ $y = x$ ○ $y = x^2$ ○ $y = x^3$ ○ $y = \sqrt{x}$ ○ $y = \lvert x \rvert$ ○ $y = 2^x$ ○ $y = \frac{1}{2}^x$	**Recommended Scoring** **(3 Total Points)**	Partial credit is not recommended.

Question	4a	Benchmark(s)	MA.912.F.2.1
a. The transformation of $H(x)$ represented by $M(x) = -x^2$ can be shown by ○ adding -1 to y-values of the function $H(x)$. ○ adding -1 to x-values of the function $H(x)$. ○ multiplying the y-values of the function $H(x)$ by -1. ○ multiplying the x- and the y-values of the function $H(x)$ by -1.		**Recommended Scoring** **(4 Total Points)**	Partial credit is not recommended.

Question	4b	Benchmark(s)	MA.912.F.2.1
b. The transformation of $H(x)$ represented by $N(x) = (x-5)^2$ can be shown by ○ adding 5 to x-values of the function $H(x)$. ○ adding -5 to x-values of the function $H(x)$. ○ subtracting 5 from x-values of the function $H(x)$. ○ subtracting 25 from x-values of the function $H(x)$.		**Recommended Scoring** **(4 Total Points)**	Partial credit is not recommended.

Question	4c	Benchmark(s)	MA.912.F.2.1
		**Recommended Scoring** **(4 Total Points)**	Partial credit is not recommended.

<table>
<tr><th>Question</th><th>5a</th><th>Benchmark(s)</th><th>MA.912.FL.3.4</th></tr>
<tr><td colspan="2">a. Complete the statements.

Savings Account B is an example of [○ linear / ○ exponential] growth because [○ it has more money. / ○ the interest earned each year is constant. / ○ the interest earned each year is compounded.] The equation that can be used to determine the amount of money in Savings Account A is [○ $A = P(1 + rt)$, / ○ $A = P\left(1 + \frac{r}{n}\right)^{nt}$,] where A = final amount, P = principal or original amount, t = number of years, and n = number times per year the interest is being compounded, and r = rate of interest per year.</td><td>Recommended Scoring

(6 Total Points)</td><td>Consider giving students 3 points for the first statement (two answer boxes) and 3 points for the second sentence (one answer box).</td></tr>
<tr><th>Question</th><th>5b</th><th>Benchmark(s)</th><th>MA.912.FL.3.4</th></tr>
<tr><td colspan="2">

Account	Work	Amount in the Account after 21 years.
A	$730(1 + 0.06)^{21}$	\$2481.68
B	$1500(1 + 0.06(21))$	\$3,390.00

</td><td>Recommended Scoring

(8 Total Points)</td><td>Consider giving students 4 points for each total. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider a minor deduction of points.</td></tr>
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Question	6a	Benchmark(s)	MA.912.F.1.8
○ linear ○ quadratic ○ exponential		**Recommended Scoring** **(3 Total Points)**	Partial credit is not recommended.
Question	**6b**	**Benchmark(s)**	MA.912.F.1.8
○ linear ○ quadratic ○ exponential		**Recommended Scoring** **(3 Total Points)**	Partial credit is not recommended.
Question	**7a**	**Benchmark(s)**	MA.912.AR.1.1
The number of years since 2007 it takes for the number of people ages 6 to 17 who listed running as a hobby to reach the maximum amount.		**Recommended Scoring** **(4 Total Points)**	Consider each student's explanation when determining partial credit.
Question	**7b**	**Benchmark(s)**	MA.912.AR.1.1
The maximum amount, in tens of thousands, of people ages 6 to 17 who listed running as a hobby between the years 2007 and 2014.		**Recommended Scoring** **(4 Total Points)**	Consider each student's explanation when determining partial credit.

Question	8	Benchmark(s)	MA.912.F.2.3
$g(x) =$ [○ $kf(x)$, ○ $f(kx)$, ○ $f(x+k)$, ○ $f(x)+k$,] where $k =$ [○ -3. ○ $-\frac{1}{3}$. ○ $\frac{1}{3}$. ○ 3.]		**Recommended Scoring** **(4 Total Points)**	Partial credit is not recommended.
Question	**9a**	**Benchmark(s)**	MA.912.F.3.1
$r(x) = 6x^2 - 20x + 3$		**Recommended Scoring** **(3 Total Points)**	Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider a minor deduction of points.
Question	**9b**	**Benchmark(s)**	MA.912.F.3.1
$w(x) = -21x^2 - 14x + 112$		**Recommended Scoring** **(3 Total Points)**	Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider a minor deduction of points.
Question	**9c**	**Benchmark(s)**	MA.912.F.3.1
$m(x) = \frac{k(x)}{n(x)}$		**Recommended Scoring** **(4 Total Points)**	Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider a minor deduction of points.